Set Equality & Quantification

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CS 0220 2021

May 28, 2021

Overview

Proof by Cases (1.7)

Predicate Formulas (3.6)

Fact about groups of people

Any two people have either met or not.

Given a group of people G, if all pairs of people in G have met, we'll call it a *club*. If no two people in G have met, we'll call them *strangers*.

Theorem. Every collection of 6 people includes a club of 3 people or a group of 3 strangers.

Does that seem true? Try some examples on the board.

Proof (Part 1)

The proof is by case analysis. Let x denote one of the six people. Let $R = G - \{x\}$ be the rest. There are two cases:

1. Among R, at least 3 have met x.
2. Among R, at least 3 have *not* met x.

At least one of these cases must hold. Since $|R|$ is odd, either more than half in R know x or less than half in R know x (and therefore more than half do not know x).

Case 1: At least 3 have met x. Let $J \subseteq R$ be those individuals. Two subcases:

1.1 No pair in J have met each other. So, J is a group of at least 3 strangers and the theorem holds in this subcase.

1.2 Some pair in J have met each other. That pair and x are a club of 3 people and the theorem holds in this subcase, too.

That covers Case 1!

Proof (Part 2)

Case 2: At least 3 have not met x. Let $J \subseteq R$ be those individuals. Two subcases:

2.1 Every pair in J have met each other. So, R is a club of at least size 3 and the theorem holds in this subcase.

2.2 Some pair in J haven't met each other. That pair and x are a group of strangers of 3 people and the theorem holds in this subcase, too.

That covers Case 2! It's kind of the inverse-video version of Case 1.

Since we showed that only these two cases can occur and the theorem holds in both, the theorem *always* holds.

Quantifiers, Revisited

Always True (universal quantification)
$\forall x \in \mathbb{R}, x^2 + 1 \geq 0.$

- For all $x \in D$, $P(x)$ is true.
- $P(x)$ is true for every x in the set D.

Sometimes True (existential quantification)
$\exists x \in \mathbb{Z}, x$ is even and x is prime.

- There is an $x \in D$ such that $P(x)$ is true.
- $P(x)$ is true for some x in the set D.
- $P(x)$ is true for at least one $x \in D$.

Mixing quantifiers

Theorem (sparse squares): There's a perfect square arbitrarily far from its closest perfect square.

Clear? Maybe a tad vague. True? How say in math?

$\forall d \in \mathbb{N}, \exists i \in \mathbb{N}, \forall j \in \mathbb{N}$, i is a perfect square AND $|i - j| \leq d$ IMPLIES j is NOT a perfect square.

The expressions nest inside each other. The order matters.

You can think of it like a little game. I'm claiming that you can pick any d you want. I'll then pick an i that's a perfect square AND no matter what j you pick that is within d values of i, j won't be a perfect square.

So, what's my winning strategy?

Any ambiguity is too many

"If you can juggle any object, you've got a talent."

1. If $\exists o$, you can juggle o, then you've got a talent.
2. If $\forall o$, you can juggle o, then you've got a talent.

"...statistics show that, in New York, a man is mugged every 11 seconds. I would now like you to meet that man. His name is Jesse Donnally."

1. $\forall t, \exists m$, m mugged in 11 second interval t
2. $\exists m, \forall t$, m mugged in 11 second interval t

From my files

Addressing a group: "Send me all of your papers."

- $\forall x$ in group, $\forall$ papers p, x wrote p IMPLIES send(x)
- $\forall$ papers p, ($\forall x$ in group, x wrote p) IMPLIES send(x)

About a medical side effect: "Everything tastes the same"

- $\forall x, \forall y$, taste(x, now) $=$ taste(y, now)
- $\forall x$, taste(x, now) $=$ taste(x, then)

"The whole article is not available."

- not $\forall$ article part x, x is available
- $\forall$ article part x, x is not available

DeMorgan returns: Negating quantifiers

These two statements are equivalent:

- Not everyone likes chocolate.
- There's someone who doesn't like chocolate.

not $\forall x, P(x)$ is equivalent to $\exists x$, not $P(x)$.

Assertion about predicates

$\exists x, \forall y, P(x, y)$ IMPLIES $\forall y, \exists x, P(x, y)$.

If $\exists x, \forall y, P(x, y)$, there must be some specific x^* such that $\forall y, P(x^*, y)$. As a result, $\forall y, \exists x, P(x, y)$ because we can always choose x^* to be the selected x.

On the other hand,
$\forall y, \exists x, P(x, y)$ IMPLIES $\exists x, \forall y, P(x, y)$
is not true.

	y_1	y_2	y_3
x_1	T	F	F
x_2	T	T	T
x_3	F	T	F

	y_1	y_2	y_3
x_1	T	F	F
x_2	T	F	T
x_3	F	T	F